

Practice Test 2: Solutions

Total: 20 marks. Open this guide after attempting the student paper.

Grade observable mathematical and representational work. Alternative mathematically valid notation is acceptable. Follow-through credit applies to a clear method after a minor earlier error. Each criterion below belongs to exactly one scored subpart; the main total is 20.

Question 1. (2)

(a) [1] A symmetric, approximately flat average magnitude for $-20\pi \leq \omega \leq 20\pi$ rad/s, approximately zero outside. Zero mean sets the DC coefficient to zero; an isolated notch at zero is acceptable, and is not required in a schematic envelope. **0.5:** flat in-band envelope and near-zero outside on a labelled angular-frequency axis. **0.5:** both cutoffs $\pm 20\pi$ rad/s. An axis in Hz with ± 10 is a correct representation but loses the requested angular-frequency conversion half-mark.

(b) [1] The available higher frequencies produce faster temporal variation; the RMS remains 0.5, so typical overall amplitude stays comparable. Individual peaks need not be identical, and fixed RMS does not impose a fixed peak bound. **0.5:** bandwidth-to-faster-variation mechanism. **0.5:** fixed RMS with a correct amplitude interpretation. Do not require a claim about exact maxima or plateau height.

Question 2. (2)

For subthreshold evolution, $\dot{v} = (J - v)/\tau_{RC}$. The currents are $J_+ = 3/2$, $J_- = 1/2$.

Positive-encoder trace [1]: initially rises concavely toward $3/2$, crosses threshold 1, emits a spike, resets to zero, stays at zero for 2 ms, then repeats. **0.5:** threshold-reaching rising trajectory for the correct current; **0.5:** spike, reset, and nonzero refractory hold. Exact curvature is qualitative, not a fitted exponential.

Negative-encoder trace [1]: rises concavely toward $1/2$, below threshold, with no spikes or repeated resets. **0.5:** correct asymptote; **0.5:** no threshold crossing/spiking and labelled voltage/time/threshold. Pure conclusions without traces receive at most one total mark. $v_-(t) = \frac{1}{2}(1 - e^{-t/\tau_{RC}})$ is consistent; a correct formula can support, but does not replace, the requested sketch.

Question 3. (7)

(a) [2]

$$H(\omega) = \frac{\langle X(\omega)R^*(\omega) \rangle}{\langle |R(\omega)|^2 \rangle}, \quad \hat{X}(\omega) = H(\omega)R(\omega).$$

At frequencies with zero response power, the ratio is undefined; one can set the unused filter coefficient to zero. The filter is optimal among unconstrained linear filters for the assumed ensemble statistics, and need not be causal.

1: correct averaged cross-spectrum divided by averaged response power, including conjugation. A single-record ratio without averaging receives 0.5 here. **0.5:** aligned sketches distinguish bandlimited input/output from a response extending beyond the signal band. **0.5:** mechanism: outside the input band, input-response cross-power vanishes under the stated assumption despite nonzero response power,

so the ideal filter suppresses the unrelated response components. Accept finite-sample residual noise and imperfect in-band decoding; do not demand that the output spectrum exactly equals the input spectrum or that the optimal filter is a perfect rectangle. Saying only “low-pass removes noise” earns the sketch half-mark if correctly drawn, but not the mechanism half-mark.

(b) [3]

$$c_0 = \int_0^\infty e^{-t/\tau} dt = \tau, \quad c_1 = \int_0^\infty t e^{-t/\tau} dt = \tau^2.$$

For example, integration by parts gives $c_1 = [-\tau t e^{-t/\tau}]_0^\infty + \tau \int_0^\infty e^{-t/\tau} dt = \tau^2$. Thus $h_0 = \tau^{-1} e^{-t/\tau}$, $h_1 = t \tau^{-2} e^{-t/\tau}$ for nonnegative time. c_n has units s^{n+1} and both normalized kernels have units s^{-1} .

1: $c_0 = \tau$ with valid integral/evaluation; **1:** $c_1 = \tau^2$ with valid integral/evaluation (0.5 each if only the final constant is correct). **1 sketch: 0.5** for causal zero-before-onset traces, $h_0(0) = 1/\tau$ with immediate decay and peak at 0; **0.5** for $h_1(0) = 0$, rise and decay with peak at τ . Label time and kernel height. Exact peak height $1/(e\tau)$ is welcome but not required. Unit area does not mean equal peak heights. A peak-at-zero error after a correct derivative deserves follow-through only for other correct features.

(c) [2] **1** for the two tau effects with mechanism: broadening the unit-area kernel averages spikes over a longer interval, reducing rapid jitter/high-frequency fluctuations (**0.5**); its causal weighting produces more temporal lag and attenuation of fast signal variation (**0.5**). For $n = 0$, $H(\omega) = 1/(1 + i\omega\tau)$; increasing $\omega\tau$ reduces magnitude and increases phase lag. The formula is optional supporting work.

1 for the bandwidth comparison: the 2 Hz signal concentrates power where the fixed filter has gain nearer one and smaller phase distortion (**0.5**), so it suffers less smoothing distortion/lag relative to its timescale than the faster input (**0.5**). The common decoder is appropriate only within the represented/trained range and assuming comparable amplitude statistics. Residual spike noise and representation error remain; reduced bandwidth need not guarantee a strictly lower measured total RMSE for every realization. A bare “2 Hz is smoother” without filter interaction earns at most 0.5 of this comparison mark.

Question 4. (3)

(a) [2]

$$D^T = (AA^T + N\sigma^2 I_n)^{-1} AX^T, \quad D \in \mathbb{R}^{d \times n}.$$

Equivalently $D = XA^T(AA^T + N\sigma^2 I_n)^{-1}$. **0.5:** compatible activity covariance AA^T and identity of size n ; **0.5:** regularizer $N\sigma^2 I_n$; **0.5:** cross-term AX^T with correct inverse/orientation; **0.5:** $d \times n$ decoder shape. A normal equation $D(AA^T + N\sigma^2 I_n) = XA^T$ is fully valid if the solved decoder is clear. Positive penalty ensures invertibility; if $\sigma = 0$, an inverse requires full row rank or a pseudoinverse.

(b) [1]

$$s_{ik} = \sum_{\ell \leq k} b_{i\ell} h_{k-\ell}, \quad \hat{\mathbf{x}}_k = D \mathbf{s}_k.$$

There is no additional Δt in the first sum. Counts are dimensionless; s has rate units. In the impulse-height convention $a_{i\ell} = b_{i\ell}/\Delta t$, the Riemann convolution is $s_{ik} \approx \Delta t \sum_{\ell \leq k} a_{i\ell} h_{k-\ell}$, giving the same count formula. **0.5:** correctly shifted spike/kernel sum; **0.5:** correct factor decision and cancellation explanation. An equivalent “add one copy of h per spike” expression is acceptable. An extra Δt with this count convention is incorrect. Continuous normalization is assumed; very coarse/truncated discrete kernels may require renormalizing their discrete area.

Question 5. (6)

(a) [2]

$$J_i = \alpha_i \mathbf{e}_i^T D^f \mathbf{a} + J_i^{\text{bias}} = \sum_{j=1}^n \left(\alpha_i \sum_{k=1}^q e_{ik} D_{kj}^f \right) a_j + J_i^{\text{bias}}.$$

Hence $w_{ij} = \alpha_i \sum_{k=1}^q e_{ik} D_{kj}^f$, and with $E_{ik} = \alpha_i e_{ik}$,

$$E \in \mathbb{R}^{m \times q}, \quad W^f = ED^f \in \mathbb{R}^{m \times n}.$$

1: substituting the decoder in the encoding equation and identifying the coefficient of presynaptic activity (0.5 for a correct current expression, 0.5 for the component weight). **0.5:** gain-scaled matrix product. **0.5:** correct matrix dimensions/orientation. Bias is separate and is added once per postsynaptic neuron. A common unit-area synaptic filter can be included before postsynaptic neural nonlinearity; the rate derivation requested here needs no filter.

(b) [1] Desired $y = 1 + \sin(2\pi t)$ spans $[0, 2]$. **0.5:** range. **0.5:** for the positive-sine half-cycle, the target exceeds B's trained radius, so accuracy is unsupported there; distortion/bias may be greater than ordinary within-range errors. Accept a proposed larger radius/retraining explanation. Do not require literal hard clipping: LIF tuning and decoder extrapolation do not generally enforce a hard bound. Dynamic filtering also adds lag, but lag alone does not address the trained-range limitation.

(c) [3] One valid pair of diagrams is

```
A_x (1-D x) -- decode x/2 --\
                                > B_z (1-D z) -- identity readout --> z
A_y (1-D y) -- decode 2y ---/

A_x (1-D x) -- embed (x,0) --\
                                > C (2-D (x,y)) -- decode xy --> B_p (1-D p)
A_y (1-D y) -- embed (0,y) --/                               [or direct p readout]
```

1: additive diagram has two appropriately transformed branches, summed into a scalar target. **1:** multiplication diagram has a joint 2-D representation, correctly injected coordinates, and a scalar product function readout/connection. This can also be drawn with the external inputs directly feeding the joint population, provided the original scalar input representations and routing are clear.

1: mechanism distinguishing architectures: independent decoded branches followed only by linear summation produce a separable function $F(x) + G(y)$; xy requires interaction between x and y , which is available through nonlinear tuning to their joint coordinates. A concise proof is valid: $F(x) + G(y)$ has zero mixed finite difference, while at the corners $(0,0), (1,0), (0,1), (1,1)$, xy has mixed difference 1. Alternatively, the effect of changing x on xy depends on y , whereas changing x in $F(x) + G(y)$ cannot depend on y . “Multiplication is nonlinear” alone earns at most 0.5 here. Dendritic nonlinear computation is a valid source-supported alternative if explicitly introduced and explained; it falls outside the question’s ordinary linear-summation restriction. A nonlinear readout from B after B represents only the one-dimensional sum generally lacks enough information to recover arbitrary xy .